INORGANIC CUME

Inorganic Chemistry of First-Row Transition Metals

Oct-2-2009

1). Cr is often used in making alloys with other metals. Give one example of the alloys that you would use daily. (10')

2). Ti is also often used in making alloys. Give one example for the alloys that would show shape-memory effect. (10')

3). Describe the trend for melting points of the first-row transition metals oxides. Point out which oxide would give the highest melting point. (20')

4). Among the first-row transition metal oxides, which one(s) would be biocompatible and which one(s) would be amphoteric? (20')

5). Define a simple chemical reaction for making ferrocene. What is the ground state spectroscopic term for the metal in the ferrocene? (20')

6). One undergraduate student came to report to you that he yesterday dissolved $Fe(NO_3)_2$ in two bottles of aqueous solutions, one was 1.0 M sodium chloride and another one was 1.0 M sodium cyanide, and he forgot to label the two bottles. How would you help the student quickly find out which is which, if you would have an EPR (or ESR) near you? Please give details in your reasoning. (20')